Impulse Is The Change Of Momentum

Synopsis: Momentum, p, measured in kg m/s, is a fundamental physical quantity, as fundamental as matter and energy. You learned that just as matter and energy's total always remains the same, i.e., conserved, the total momentum always remains the same. So, if you add up the momentum of every single matter in the universe, this total momentum of the universe will remain constant. Such a calculation is likely intractable given the vast size of the universe, but the idea is clear. We can, however, define our own "universe" or, in physics lingo, "system." The system is what we represent to be the universe of interest. If we also make this universe "closed," we now have a completely contained system where everything inside this system strictly follows all conservation laws (matter, energy, and momentum). The idea of the "system" is an important topic, but for now, it suffices to say that a "closed system" is a contained universe of interest as we define it. For example, a system might be two balls in an empty space (no external forces like gravity or friction) moving toward each other along a straight line.

The primary focus of this Note is impulse. As the title states, an impulse is simply the change of momentum. That is it. So, using the usual physics notation for change, delta, this change of momentum is Δp and follows the standard convention of final minus initial.

Impulse has a symbol and unit: J [kg m/s = N s] where the second equality is (Newton second, and you should convince yourself that [N s] is indeed the same as [kg m/s]).

Impulse: $J = \Delta p = p_f - p_i$. So what about impulse? Remember, a moving massive object such as a freight train is almost impossible to stop with the strength of a typical human. It's because of the massive momentum of the freight train, and to cause the change in momentum (slow it down, for example), a very large force is required. It turns out we have a bit of a way out: This change in momentum is brought about by the applied force times the duration of this applied force. So, if you can only apply a small force, so long as you can apply this force for a very long time, you will be able to cause enough change in the momentum to slow/stop the freight train, eventually.

Therefore, another statement for impulse is: $J = F\Delta t$, where F is the applied force in N and Δt is the duration of the applied force in seconds. This way of stating impulse is actually just a variation of Newton's second law, F = ma. If you write $a = \Delta v/\Delta t$ (change of velocity over the change of time), you see that Newton's second law is actually $F = m(\Delta v/\Delta t)$, and $m\Delta v$ is none other than the change in momentum, J!

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$J = \Delta p = m\Delta v = m(v_f - v_i)$ and, therefore, $F = J/\Delta t$. Rearranging, $J = F\Delta t$.

In this Note, we will use the impulse equation to solve problems involving momentum change, i.e., collisions.

Performance Expectations: Apply the impulse equation to solve collision problems.

For PHY 151 — with calculus: Treat impulse as the integral of a varying force, $J = \int F\, dt$ — the area under a force–time graph — and define the average force as the constant force with the same area.

Instructions and Questions

1) A force, for a time.
   i) A tow-truck applies a force of 2000 N on a 2000-kg car for a period of 3 seconds. (A) What is the magnitude of the change in the car's momentum? (B) If the car starts at rest, what will be its speed after 3 seconds?
   ii) A 2-kilogram body is initially traveling at a velocity of 40 meters per second east. If a constant force of 10 newtons due east is applied to the body for 5 seconds, the final speed of the body is: 15 m/s / 25 m/s / 65 m/s / 130 m/s.

2) Stopping something: the force depends on how long you take.
   i) A motorcycle being driven on a dirt path hits a rock. Its 60-kilogram cyclist is projected over the handlebars at 20 meters per second into a haystack. If the cyclist is brought to rest in 0.50 seconds,

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the magnitude of the average force exerted on the cyclist by the haystack is: 6.0×10^{1} N / 5.9×10^{2} N / 1.2×10^{3} N / 2.4×10^{3} N.

ii) The instant before a batter hits a 0.14-kilogram baseball, the velocity of the ball is 45 meters per second west. The instant after the batter hits the ball, the ball's velocity is 35 meters per second east. The bat and ball are in contact for 1.0×10^{-2} second. Calculate the magnitude of the average force the bat exerts on the ball while they are in contact.

iii) In an automobile collision, a 44-kilogram passenger moving at 15 meters per second is brought to rest by an air bag during a 0.10-second time interval. What is the magnitude of the average force exerted on the passenger during this time? 440 N / 660 N / 4400 N / 6600 N.

3) Non-constant forces.
But not all forces are constant. What do you do if a changing force is applied for a period of time? In that case, we can make a graph of the force applied on the y-axis vs. time on the x-axis. The area under the Force–Time curve is the impulse, or change in momentum.

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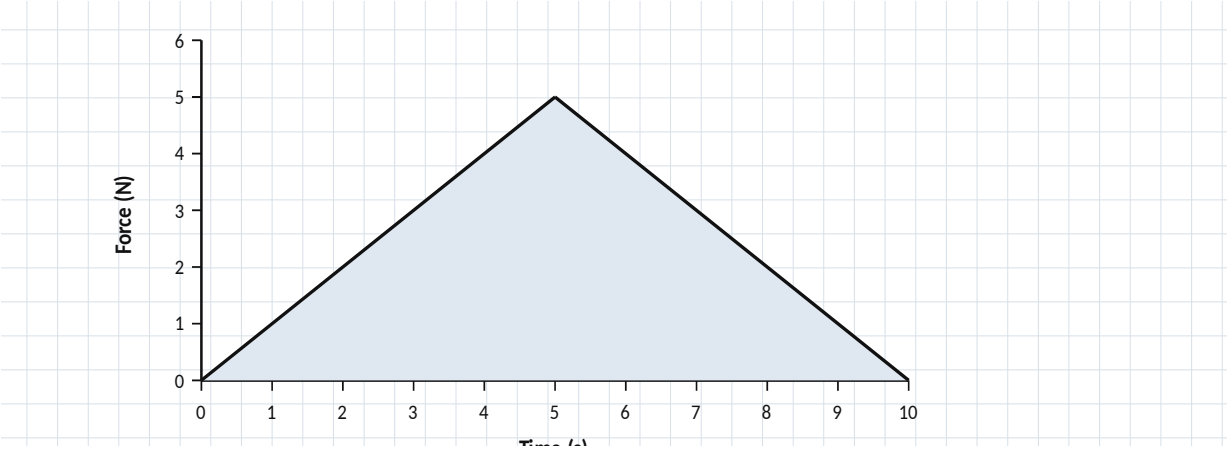

i) For the case of the sample graph, determine the impulse applied by calculating the area of the triangle under the curve.
ii) A 2.0 kg cart at rest receives that impulse. How fast is it moving afterwards?
iii) What constant force, acting for the same 10 s, would have delivered the same impulse? Mark it on the graph.

Impulse as an integral.
For a force that varies, $J = \int F\, dt$ over the contact, which is exactly the area under the force–time graph — the triangle above is the case where you can do the integral by geometry. The average force is defined by $F_{avg}\, \Delta t = J$: the constant force that would have delivered the same impulse in the same time. A force measured by a sensor during a real collision is a spike, often ten times the average; the area is what the momentum feels.

i) A force $F(t) = 60\, t - 6\, t^2$ N acts from t = 0 to t = 10 s (it starts and ends at zero, peaking at 150 N). Integrate to find the impulse, and the average force over the 10 s.
ii) The baseball in section 2 felt an average of 1.1×10^3 N over 0.010 s. If the real force was a triangular spike over the same 0.010 s, what was the peak force?

4) Looking ahead.
Impulse is the single-body side of the conservation law: what one cart does to the other during contact is the same $F\Delta t$ with opposite signs. Next, the last question of the module — what happens to the kinetic energy in a collision, and why some collisions keep it and some do not.

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